ITC C503 DATA ANALYTICS MIDTERM PROJECT

INSTRUCTIONS:

MIDTERM PROJECT INSTRUCTIONS

Instructions

Solve all the problems provided in the attached file. Follow the instructions below for your submission:

TAKE NOTE: PROJECT compose of PART I and PART II given on the succeeding pages

1. **Use a Spreadsheet Application**
    - Use **Microsoft Excel** or **Google Sheets** to perform all necessary computations.
    - Use appropriate spreadsheet formulas for each problem.
2. **Format Your Answers**
    - Add appropriate **labels** to your table and answers.
    - Apply appropriate **font colors** to distinguish important information.
    - Place a **box/border** around your final answers.
3. **Screenshot of Spreadsheet Table and Answers**
    - Take a **screenshot of the completed table and answer for each problem**.
    - Make sure the problem number and the corresponding answer are clearly visible.
4. **Prepare One Document**
    - Compile all screenshots in **one document**.
    - Arrange the screenshots **according to problem number**.
    - For each problem, include:
        - Spreadsheet data/table
        - Computation and final answer
        - Spreadsheet formulas used

5. **Document Header and Footer**

   Header: **ITC C503 DATA ANALYTICS MIDTERM PROJECT**

   Footer: **Prepared By: Your Complete Name and Section**

6. **Save Your File**

   Save your document using the following filename format:

   ITC C53_MidtermProject_LastName

7. **Convert to PDF**

   - Convert your completed document into a **PDF file**.
   - Review the PDF to ensure that all tables, screenshots, formulas, and answers are clearly visible and properly arranged.

8. **Submission**

   - Submit the **PDF file** containing your complete Midterm Project.
   - Make sure your **name, section, problem numbers, computations, formulas, and final answers** are complete before submitting.

PART I – GROUPED FREQUENCY DISTRIBUTION TABLE AND STATISTICAL CHARTS

1. The number of counties, divisions, or parishes for each of the 50 states is given below.

Counties, Divisions, or Parishes for 50 States

67	27	15	75	58	64	8	67	159	5
102	44	92	99	105	120	64	16	23	14
83	87	82	114	56	93	16	10	21	33
62	100	53	88	77	36	67	5	46	66
95	254	29	14	95	39	55	72	23	3

Source: *World Almanac and Book of Facts.*

	A	B
1	Raw Data	Organized Data
2	67	3
3	27	5
4	15	5
5	75	8
6	58	10
7	64	14
8	8	14
9	67	15
10	159	16
11	5	16
12	102	21
13	44	23
14	92	23
15	99	27
16	105	29
17	120	33
18	64	36
19	16	39
20	23	44
21	14	46
22	83	53
23	87	55
24	82	56
25	114	58
26	56	62
27	93	64
28	16	64
29	10	66
30	21	67
31	33	67
32	62	67
33	100	72
34	53	75
35	88	77
36	77	82
37	36	83
38	67	87
39	5	88
40	46	92
41	66	93
42	95	95
43	254	95
44	29	99
45	14	100
46	95	102
47	39	105
48	55	114
49	72	120
50	23	159
51	3	254

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1.1 Create a Grouped Frequency Table

- Use the data to construct a grouped frequency distribution with 6 classes
- Determine the frequency of each class limits

Frequency
19
17
12
1
0
1

Total Frequency
50

- Determine the class boundaries of each class limits

Class Boundaries		
Class	Lower Limits	Upper Limits
1	2.5	44.5
2	44.5	86.5
3	86.5	128.5
4	128.5	170.5
5	170.5	212.5
6	212.5	254.5

- Determine the Class Median of each class limits

Class Midpoint
23.5
65.5
107.5
149.5
191.5
233.5

- Compute cumulative frequency

Cumulative Frequency
19
36
48
49
49
50

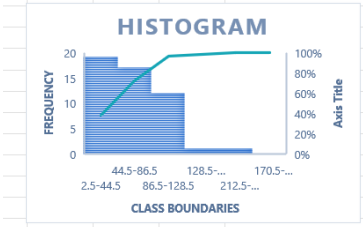
- Frequency percentage

Frequency Percentage
38%
34%
24%
2%
0%
2%

2. Use your Creeated Grouped Frequency Distribution Table on Problem No.1 to construct the following statistical chart

2.1 Histogram - Y axis - Frequency , X axis - Class Boundaries

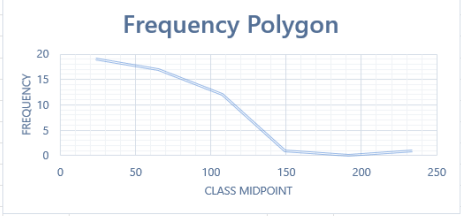
Histogram



Class Boundaries	Frequency
2.5-44.5	19
44.5-86.5	17
86.5-128.5	12
128.5-170.5	1
170.5-212.5	0
212.5-254.5	1

2.2 Frequency Polygon - Y axis - Frequency , X -axis Class Midpoints

Frequency Polygon

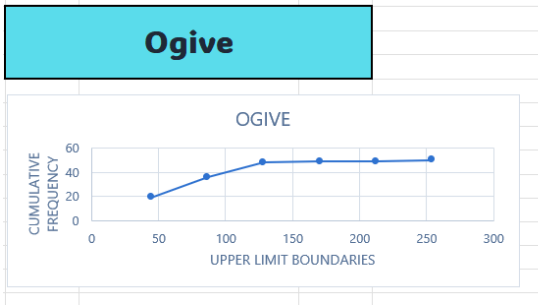


Class Midpoint	Frequency
23.5	19
65.5	17
107.5	12
149.5	1
191.5	0
233.5	1

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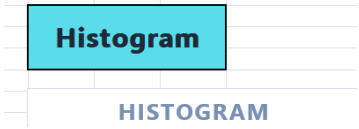
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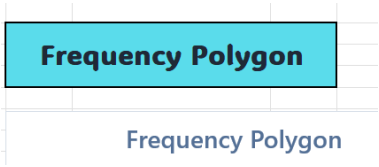
2.3 Ogive - Y-axis Cumulative Frequency , X - Axis Class Boundaries



Upper Limit Boundaries	Cumulative Frequency
44.5	19
86.5	36
128.5	48
170.5	49
212.5	49
254.5	50

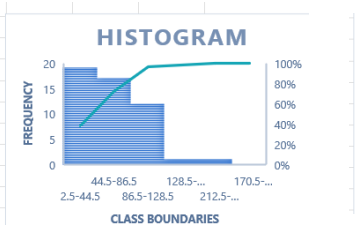
2.4 ADD Data Labels on each chart

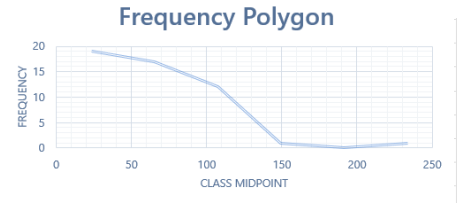


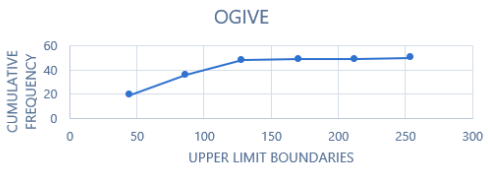




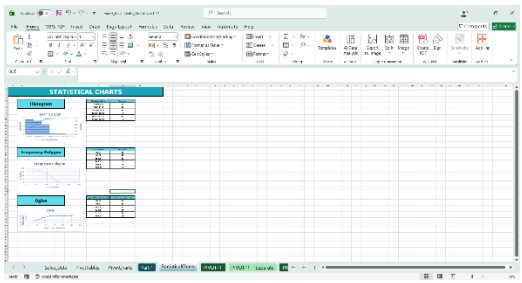
2.5 ADD Chart Titles, Axis Titles (X,Y) , Series Name







2.6 On New Sheet Tab Move your 3 Charts Created



2.7 Rename the New Sheet Tab to StatisticalCharts

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StatisticalCharts

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PART II – PIVOT TABLE AND PIVOT CHART

- Down load the Attached Excel File – Excel Pivot Table Dashboard
- Create a Pivot Table, follow the Dashboard Guide given on PivotTable sheet tab
  - Add Title on each Pivot Table



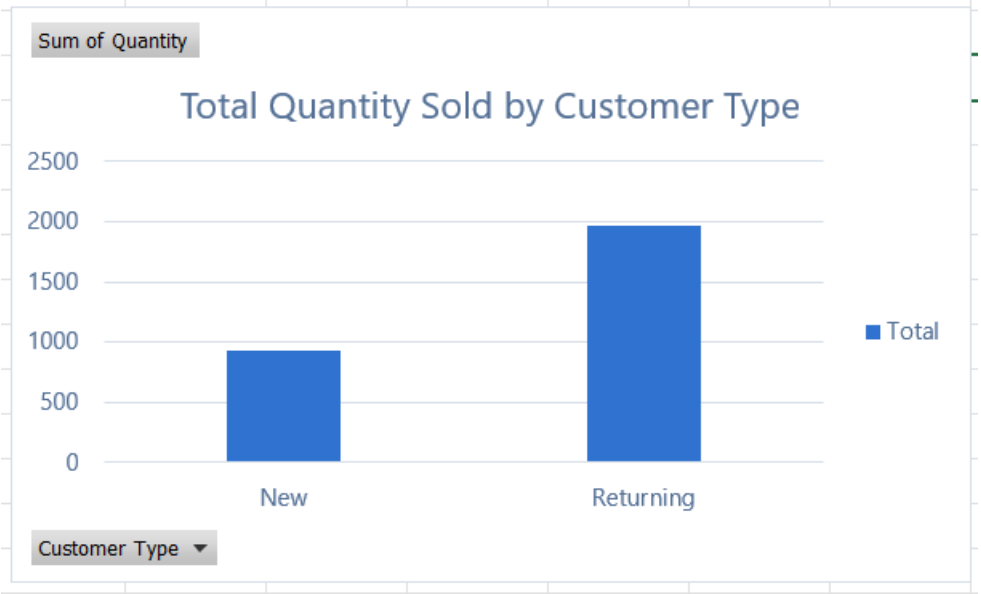
- Create a Pivot chart on each created Pivot Table
  - Add Title and Legend on each chart created



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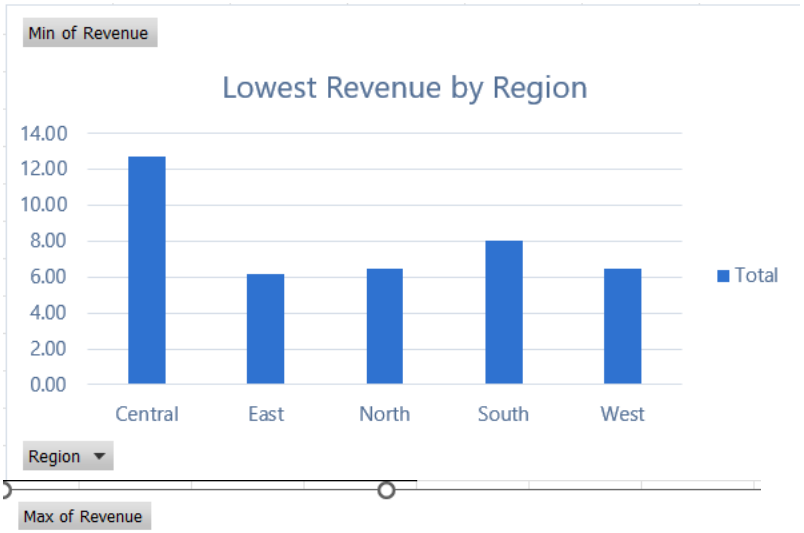
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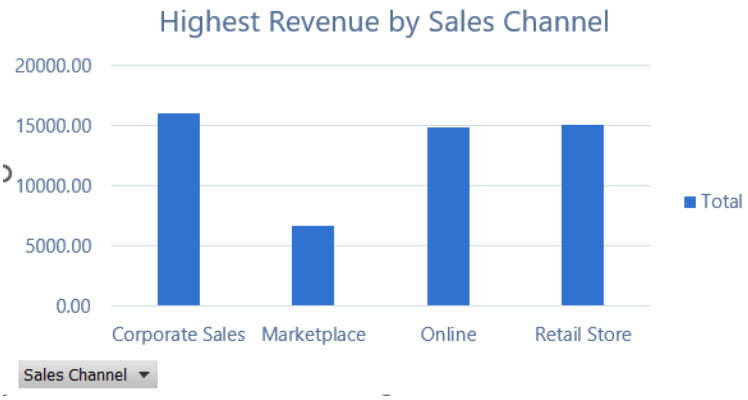






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- Add Data Labels on each chart created
  - **DONE**
- Display SELECT ALL option on each chart created
  - **DONE**
- Display FILTERED option on each chart created
  - **DONE**